

Generation Interconnection System Impact Study Report

For Queue Project Q21-1781

New Salisbury 138 kV

February 19, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1781. The Interconnection Customer proposes a 306.1 MW wind facility interconnecting at the New Salisbury 138 kV substation in the Virginia area. The request was received on June 13, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 482 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$30,658,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: BLACKBERRY - NEW SALISBURY 138 KV
Contingency: NEW SALISBURY - TIMBERVILLE 138 KV line
Loading: 160.2% (Rating: 285 MVA)
DFAX: 100.0%
Pre-project loading: 52.7%
2. Facility: CHATOM - BLACBERRY 138 KV
Contingency: NEW SALISBURY - TIMBERVILLE SUBSTATION 138 KV
Loading: 155.2% (Rating: 290 MVA)
DFAX: 100.0%
Pre-project loading: 49.7%
3. Facility: MIDDLETOWN JUNCTION - MOUNT PLEASANT 138 KV
Contingency: MOUNT PLEASANT - FT BENTON 138 KV
Loading: 134.5% (Rating: 220 MVA)
DFAX: 5.0%
Pre-project loading: 127.4%
4. Facility: CHATOM - ALVIN 138 KV
Contingency: PAGE - PURDY 138 KV
Loading: 126.5% (Rating: 397 MVA)
DFAX: 23.8%
Pre-project loading: 108.2%
5. Facility: MOUNT PLEASANT - FT BENTON 138 KV Line
Contingency: FT BENTON - MIDDLETOWN JCT 138 KV
Loading: 122.6% (Rating: 229 MVA)
DFAX: 7.2%
Pre-project loading: 112.9%
6. Facility: S JORDAN - MOUNT PLEASANT 138 KV
Contingency: MOUNT PLEASANT - FIREBANTZ 138 KV
Loading: 119.9% (Rating: 232 MVA)

DFAX: 5.8%

Pre-project loading: 112.2%

7. Facility: FT BENTON - MIDDLETOWN JCT 138 KV

Contingency: FT BENTON - MOUNT PLEASANT 138 KV

Loading: 118.5% (Rating: 219 MVA)

DFAX: 7.2%

Pre-project loading: 108.5%

8. Facility: ALVIN - NUCLEAR FUELS 138 KV

Contingency: PAGE - PURDY 138 KV

Loading: 107.9% (Rating: 475 MVA)

DFAX: 23.8%

Pre-project loading: 92.6%

9. Facility: ST LOUIS - NICHOLS 138 KV

Contingency: PAGE - PURDY SUBSTATION 138 KV line

Loading: 102.1% (Rating: 512 MVA)

DFAX: 18.4%

Pre-project loading: 91.1%

10. Facility: S BELLEVILLE - NICHOLS 138 KV

Contingency: PAGE - PURDY 138 KV Line

Loading: 102.0% (Rating: 506 MVA)

DFAX: 18.4%

Pre-project loading: 90.9%

11. Facility: OAK CRK STA - SANS SOUCI 138 KV

Contingency: NCLEAR FUELS - ALVIN 138 KV

Loading: 100.8% (Rating: 372 MVA)

DFAX: 8.2%

Pre-project loading: 94.1%

12. Facility: HARRISONBURG - SANS SOUCI 138 KV Line

Contingency: NUCLEAR FUELS SUBSTATION - ALVIN 138 KV L

Loading: 100.8% (Rating: 372 MVA)

DFAX: 8.2%

Pre-project loading: 94.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BLACKBERRY - NEW SALISBURY 138 KV	Rebuild line with higher-capacity conductor	\$6,053,000	66% (\$4,011,000)
CHATOM - BLACBERRY 138 KV	Rebuild line with higher-capacity conductor	\$10,648,000	66% (\$7,056,000)
MOUNT PLEASANT - MIDDLETOWN JUNCTION 138 KV	Rebuild line with higher-capacity conductor	\$10,417,000	20% (\$2,128,000)
ALVIN - CHATOM SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$4,740,000	20% (\$963,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MOUNT PLEASANT - FT BENTON 138 KV Line	Rebuild line with higher-capacity conductor	\$34,756,000	22% (\$7,574,000)
SOUTH JORDAN - MOUNT PLEASANT SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$20,053,000	22% (\$4,341,000)
FT BNTON - MIDDLETOWN JCT 138 KV	Rebuild line with higher-capacity conductor	\$10,681,000	23% (\$2,433,000)
NUCLEAR FUES - ALVIN 138 KV	Reconductor line	\$3,411,000	20% (\$693,000)
ST LOUIS - NICHOLS 138 KV	Reconductor line	\$2,086,000	20% (\$411,000)
S BELLEVILLE - NICHOLS 138 KV	Reconductor line	\$3,082,000	20% (\$608,000)
SANS SOUCI - OAK CRK STATON SUBSTATION 138 KV	Reconductor line	\$994,000	24% (\$237,000)
HARRISONBURG - SANS SOUCI 138 KV	Reconductor line	\$851,000	24% (\$203,000)
	Total allocated to project		\$30,658,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.